

**CSE7101- Capstone Project
Review-2**

Smart Curriculum Activity & Attendance App

Batch Number: CSE_109

Roll Number	Student Name
YASHAS RAJ S	20231CSE3019
RAKSHITHA REDDY KS	20231CSE3089
SACHIN KUMAR SAROJ	20231CSE3018

Under the Supervision of,

**Dr.ChandraSekhar M
School of Computer Science and Engineering
Presidency University**

Name of the Program: EduTrack

Name of the HoD: Dr. Asif Mohammad

Name of the Program Project Coordinator: Dr.ChandraSekhar M

Name of the School Project Coordinators: Dr. Sampath A K , Dr. Geetha A



Problem Statement Number: SIH25011

Smart Curriculum Activity & Attendance App

Category (Hardware / Software / Both): Software

Problem Description:

Develop an integrated mobile/web application that automates curriculum activity tracking and student attendance management. The system should provide real-time monitoring of student participation in academic and co-curricular activities, generate comprehensive attendance reports, and facilitate better academic planning and student engagement tracking. The solution should be scalable, user-friendly, and integrate with existing educational management systems.

Project Object:

- Automate Student Attendance Management
- Digitally Track Curriculum Activities
- Improve Academic Transparency and Accountability
- Centralize Academic Data Management
- Enable Data-Driven Academic Decision Making
- Support Data-Driven Academic Decision Making
- Improve Faculty Efficiency and Time Management

Content

- Title slide content
 - Problem Objectives
 - Github Link
 - Chosen Solution Approach
 - Case Study
 - References
- 
- A solid blue triangle is located in the bottom-left corner of the slide, pointing towards the bottom-right.

Project Objectives

Our core objectives for this project are to:



Automate Attendance

Streamline student attendance management.



Track Activities

Digitally monitor curriculum and co-curricular involvement.



Reduce Errors

Minimize manual effort and attendance discrepancies.



Centralize Data

Consolidate academic data for easy access.



Generate Reports

Enable quick and accurate reporting.

Github Link

The Github link provided should have public access permission.

Github Link

https://github.com/YashasRaj20231CSE3019/university-project-CSE_109.git



Chosen Solution Approach



Integrated Application

Unified experience across web and mobile platforms.



Mobile Application

Convenient access for students and faculty on the go.



Web Dashboard

Comprehensive management for administrators and coordinators.



Centralized Backend

Robust system for efficient data processing and storage.



Cloud Infrastructure

Ensures high availability and scalability for future growth.

Case Study 1 – AI-Enabled Automated Attendance Systems



Paper Reference

A. Singh, R. Verma, and S. Nair, *IEEE Transactions on Education*, 2024



Problem Addressed

- Manual attendance systems are time-consuming and error-prone
- Lack of automation in smart classroom environments



Proposed Solution

- AI-enabled attendance using facial recognition
- Automated verification to prevent proxy attendance



Key Outcomes

- Significant reduction in faculty workload
- Improved accuracy and reliability of attendance records



Identified Gap

- Focuses only on attendance
- No integration with curriculum activities or engagement analytics



Relevance to Our Project

- Inspires AI-based attendance automation
- Our system extends this by integrating curriculum activity tracking and reports

Case Study 2 – Mobile-Based Student Engagement Monitoring

	Paper Reference L. Anderson, M. Roberts, and T. Clark, <i>IEEE Access</i> , 2024
	Problem Addressed • Difficulty in monitoring student engagement beyond classroom attendance • Limited real-time insights into student participation
	Proposed Solution • Mobile application for engagement tracking • Learning analytics to monitor student interaction patterns
	Key Outcomes • Improved visibility into student engagement levels • Better feedback mechanisms for faculty
	Identified Gap • Does not include attendance automation • Limited predictive analytics capabilities
	Relevance to Our Project • Supports the use of mobile apps for engagement tracking • Our project combines attendance and engagement in a unified platform

Case Study 3 – ML-Based Educational Data Analytics

	Paper Reference Y. Liu, H. Zhang, and J. Kim, <i>IEEE Transactions on Learning Technologies</i> , 2024
	Problem Addressed • Lack of predictive insights into student academic performance • Reactive academic interventions instead of proactive ones
	Proposed Solution • Machine learning models for performance prediction • Analysis of attendance, engagement, and academic data
	Key Outcomes • Early identification of at-risk students • Improved academic planning and intervention strategies
	Identified Gap • Focuses on analytics, not system-level implementation • No real-time attendance or activity capture mechanism
	Relevance to Our Project • Forms the foundation for predictive analytics • Our system provides real-time data for effective ML-based insights

References (IEEE Paper format)

1. A. Singh, R. Verma, and S. Nair, “AI-enabled automated attendance management systems for smart classrooms,” IEEE Transactions on Education, vol. 67, no. 2, pp. 162–173, May 2024.
2. L. Anderson, M. Roberts, and T. Clark, “Mobile-based student engagement monitoring using learning analytics,” IEEE Access, vol. 12, pp. 45890–45904, Feb. 2024.
3. Y. Liu, H. Zhang, and J. Kim, “Machine learning-driven educational data analytics for student performance prediction,” IEEE Transactions on Learning Technologies, vol. 17, no. 1, pp. 34–46, Jan. 2024.
4. S. K. Rao, P. Mehta, and A. Banerjee, “Secure cloud-based student information systems for higher education institutions,” Journal of Educational Technology & Society, vol. 27, no. 1, pp. 201–214, Jan. 2024.
5. E. Wilson, D. Harper, and J. Moore, “Real-time academic performance monitoring using IoT and analytics,” IEEE Transactions on Systems, Man, and Cybernetics: Systems, vol. 55, no. 4, pp. 2891–2903, Apr. 2025.



Thank
You!